Shifting Hulls

A budgeted decision agent for a fleet that changes its identity

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1. Project description

Ships move sanctioned oil by changing the things that identify them. A hull keeps its IMO number for life, but its name, flag and the MMSI broadcast by its AIS transponder can be swapped at will: one tanker (IMO 9319703) appears between 2012 and 2026 under 10 MMSIs, 7 names and 8 flags, from MAERSK PIPER (Singapore) to FIANIT (Russia). Follow the MMSI: ten ships. Follow the hull: one.

We want to build a simulated port-state-control office. It ingests two independent event streams (fast, noisy AIS-like messages and slow, delayed compliance events) and must decide, every simulated day, which of N vessels to inspect given a budget of K inspections. An LLM agent makes that decision by calling smaller models and tools over MCP; the inspection list returns to the simulator, where inspected vessels adapt. It is a testbed. The claim is narrow and falsifiable: *a small, locally served agent that can query an event store catches at least two more shadow hulls per 20 inspections than the better of the churn rule and the scorer alone* (ten paired seeds; lower end of the 95% interval at or above +2); and we *measure* how fast either degrades as the fleet changes tactics.

The idea rests on a pilot with real data (Table 2): hulls tagged *shadow fleet* on the Ukrainian sanctions list (OpenSanctions, with identities from Global Fishing Watch) took new MMSIs 3.6 times faster per vessel-year than hulls merely detained in port (0.65 vs 0.18), and 59.7% took at least three new MMSIs since 2022 against 8.1%. One such count separates the groups with AUC 0.84, but 0.77 within tankers (only 23 controls), and churn is measured today, after designation, so some may follow the listing. It is a signal, not a detector; that difference is the project.

2. Problem definition

Identity is not an identifier. Only 35.0% of the 20,906 OpenSanctions vessel records carry an MMSI (65% of the 9,425 distinct hulls have one somewhere), against 96.2% with an IMO. Linking a list to what a ship broadcasts is inference over names, flags and tracks. It fails exactly where it matters: 26 shadow-tagged hulls appear by IMO in one day of US AIS, and 25 broadcast an MMSI different from the one listed today. In the pilot, 65 of 71 multi-identity shadow hulls hand over cleanly in the registry table, yet in the self-reported table 30 of 149 shadow and 23 of 149 control hulls show two MMSIs overlapping for over a month: some counted "new MMSIs" are concurrent transponders, not replacements.

Two streams, two clocks. One day of US coastal AIS is 8,156,509 messages from 15,915 vessels; the compliance list is 23,751 entities, a snapshot without event times. The simulator must inject the delay and reordering a time-keyed join would mishandle.

Measurements are noisy in ways that mimic evasion. In that day (NOAA), Draft is empty in 63.0% of records, Status in 25.1% and Cargo in 25.0%; 0.09% of moving position pairs 10–600 s apart imply over 60 km/h. The longest per-vessel transmission gap has median 12 minutes but p90 2.9 hours, and 19.5% of vessels are quiet over an hour, partly from coverage and the one-day window. A gap looks the same whether a transponder is broken or switched off.

The decision is about the top of the list. With K slots a day, AUC is the wrong metric. Simulating $N = 1000$ hulls, 5% prevalence (50 shadow) and $K = 20$, resampling pilot scores: random inspection catches 1.0 shadow hulls, the one-feature churn rule 12.5 and the gradient-boosted model 9.4 (spread across simulated fleets, 8–16 and 5–14). The model's AUC is higher (+0.045, 95% CI 0.012–0.078) yet its paired top-20 difference is −2.8 catches (−7 to +1, conditional on resampling the same 149 controls): no evidence either way. Eight of twenty slots are still wasted; that headroom is the research question. Controls are detained hulls, not a random fleet sample, so these figures are optimistic.

The target adapts. Inspections are visible. Thinning every churn feature of shadow hulls proportionally, model held fixed, cutting churn to 58% drops AUC from 0.84 to 0.74 for the one-feature rule and from 0.88 to 0.73 for the model; at 34% both are near 0.60. Subtracting a fixed number of MMSIs and flags instead leaves the model at 0.76 against 0.58 for the rule at −2, so robustness depends on how the fleet adapts. This is a counterfactual, not an observed adaptation.

Truth is thin, and time leaks. By MMSI, 99 of 15,915 US-coastal vessels match a compliance record and none is flagged; by IMO, 203 match, of which 26 are shadow-tagged. One day gives one MMSI per hull and the list reflects 2025–26, so the mismatch is partly time drift: a spot-check, not a resolver test, and too small to evaluate the agent, so evaluation runs in a simulator that knows the truth, and it must avoid the temporal bias that inflates security-ML results under careless train/test splits [1].

3. State of the art

Five strands exist separately; none we found puts an LLM agent in a budgeted loop against an adapting fleet, and we found no published ML targeting of port-state-control inspections.

The gap is a missing *closed loop* for this domain, not a missing model: Stackelberg games already treat budgeted inspection against an adversary, but not with a language-model agent over messy identity data. We claim a testbed, not novelty: an arena where "does the agent still work after the fleet adapts?" gets a number.

4. Solution

Architecture. A seeded, YAML-configured simulator emits two asynchronous streams into a message queue (Redis Streams); a consumer validates against a schema, writes raw and clean rows to PostgreSQL, quarantines rejects; an online feature service keeps per-hull identity timelines; the agent queries it through MCP tools, and a FastAPI backend serves a dashboard (ranked hulls, reasons, metrics) and a REST API; Prefect schedules the daily cycle and returns the plan to the simulator; Docker Compose runs the services; the agent is served locally (vLLM). Offline ETL replays streams

Strand	What it gives us	What it does not
Maritime intelligence, commercial [2] and ESA [3]	Vendor-described LLM agents for ownership and sanctions checks; ESA vessel-fingerprinting project	Closed; self-reported accuracy; no public data or reproducible baseline
Identity and dark-vessel science [4, 5]	Consolidates identity across 40+ registries and a decade of GPS; reflagging; AIS-off detection	Fishing fleets; offline; static analysis, not a budgeted online decision
LLM entity matching [6]	755k expert-labelled pairs; a local 14B reasoning model reaches 98.2% F1 vs 91.3% for rules	Pairwise strings; nothing below 14B; weak on transliteration; tracks not used
Agentic security operations [7, 8, 9]	Scoped-permission patterns; CaMeL proves security on 77% of AgentDojo tasks [9]; every baseline missed $\geq$40.4% of attack alerts in one scenario, the authors' multi-agent system 3.1% [8]	Static alerts, no reacting adversary; CaMeL's guarantee needs its dual-LLM split, which we only borrow in spirit
Synthetic AIS [10]; security games [12]	GAN-augmented AIS tracks; Stackelberg models of limited patrols against adaptive adversaries	No labels, identity change or compliance events; games rely on a payoff model, no LLM agent

Table 1: State of the art, by strand.

to build training sets and rerun any day.

Calibration from data. Parameters are measured: 0.65 vs 0.18 new MMSIs per vessel-year; 10.7% vs 47% of hulls with none since 2022; the gap distribution above; a 15–20% rate of overlapping MMSIs; missingness 63/25/25%. A negative binomial fits the churn histogram to total-variation distance 0.09 (shadow) and 0.03 (control), 0.15 and 0.08 on held-out halves of about 75 hulls, which is roughly the noise floor: adequate, not validated. Only this marginal is calibrated; track behaviour is not. Streams are coupled by design: a listing makes a hull's later gaps and renames more likely (a tunable parameter, not an estimate).

Models. Three roles, three metrics. (i) A resolver (4–9B LLM) decides whether a group of records is one hull and returns a confidence, given names, flags and track summaries; LLM-matching evidence is at 14B and string-level [6], so 4–9B on track-aware inputs is untested; a plain track-matching algorithm is its baseline, and the 26 IMO-keyed real hulls are its validation set. (ii) A gradient-boosted scorer outputs risk from churn, gaps and events: AUC 0.88 cross-validated, but 0.76 within tankers (rule: 0.77); port-arrival and loitering features reflect how controls were chosen (detained in port). (iii) The decision agent ($\leq$27B, 4-bit, local) sees the top candidates and the budget, calls the other two over MCP, and justifies each slot; its edge can only come from information the scorer lacks, namely free text (names, registry notes) and the stream coupling, so both are simulated. We will sweep quantization on our tool set: one published table has Qwen3 1.7–8B losing under 3 points at Q4_K_M but Gemma-3 and Phi-4-mini over 20, with nothing above 12B tested [11].

Failure and misuse. Ship names and registry notes are attacker-controlled, so retrieved text is data: the tool set is capability-scoped, only `submit_inspection_plan` mutates state, the budget is enforced server-side, and ungranted fields are unreadable [7]. This is least-privilege, not CaMeL's guarantee. On a stall or invalid plan the system falls back to the scorer's top-K.

Evaluation. In-simulator truth gives per-model metrics (resolver F1 and calibration; scorer AUC, precision@K) and one end-to-end metric (shadow hulls caught per inspection over 30 simulated days) against random, the best-threshold churn rule, the scorer alone and the agent, plus latency at 27B 4-bit. Ten seeds per condition, paired differences with intervals, and the falsification criterion in §1.

Scenarios. *Drift*: caught hulls reduce churn and shift to gaps; we measure the drop and whether re-scoring recovers it. *Load*: a 10× AIS burst; we measure queue lag and whether the plan lands within the simulated day. *Failure*: the compliance stream arrives six hours late, and separately the model server dies; the fallback must preserve precision@K.

Evidence from the pilot ($n = 298$ hulls; 149 shadow, 149 detained-only)	Value	95% CI
Mean new MMSIs since 2022, shadow vs. control	3.11 vs 0.87	–
AUC, new-MMSI count alone (flag switches 0.840, loitering 0.787)	0.837	0.789–0.881
AUC, gradient-boosted model, repeated 5-fold	0.882	0.830–0.912
tanker-only subgroup (144 vs. 23): model 0.759, rule 0.767	–	0.66–0.87 (rule)
AUC with shadow churn cut to 58%: rule 0.739, model 0.730	–	–

Table 2: Pilot evidence. Shadow group 97% tankers vs. 15% of controls.

Limitations. One label source, applied after the fact (churn may follow listing or the reverse); a ship-type confound; features measured today, not at decision time; an untrained resolver; a churn-calibrated simulator favours churn-using models.

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